

P5 "Raise Your AI Champion" — Design Proposal

Date: 2026-07-16 · **Status:** for review · **Scope:** the P5 (age 10) *project* track — 17 standalone projects, one per P5 lesson, plus the kid-facing AI buddy.

Summary

Over P5, each child raises one AI champion: a humanoid robot they design, starting as a blank shell. After each lesson, a short standalone project gives the champion one durable ability tied to that lesson's AI concept. By year-end the champion can perceive, act, plan, and judge; every ability is shown off in a shared "AI City." A personal AI buddy — which the child names — coaches them throughout and, on the child's approval, helps tune the champion.

This document covers the system: what a project is, the champion, the buddy, the 17 abilities, and the build order. Each project's detailed content and flow is specified separately.

1. Two tracks

- **Lessons** (already exist): 17 P5 games that teach the AI concepts.
- **Projects** (this proposal): 17 standalone builds where the child gives the champion an ability.

Each project is paired to a lesson **by concept only** — no shared code or data. A project re-introduces its own concept and has the child do the real work inside it, so it never depends on the lesson being played first.

2. What a project is

A project = configure and tune a pre-built module. No typing code.

The child supplies the content that matters — photos, rules, thresholds, judgment calls — in a simple editor. The engine is pre-built, so every child reaches a working champion. Skill shows in the decisions.

Student content first:

- Pre-built = the invisible engine only (classifier, sim, meters). No pre-made content.
- The child supplies all the real content, from the real world. Imperfect results are the lesson, not failure.
- The buddy guides the child to do the real job. It never supplies the content.

Cluster	The child's real-world job
Senses	Photograph / record real things
Body	Design their own course or strategy
Planner	Survey real data, design their own city
Conscience	The AI learns their own judgments

Every project runs the same six beats: **Hunt** (gather real content) → **Teach** (feed it in) → **Come alive** (the champion reacts, a part lights up) → **Prove** (test on new input) → **Show off** (in the City) → **Keep** (the ability stays).

3. The champion

- A **humanoid robot**, so every ability has a natural home (eyes, legs, heart, brain-core) and the child bonds with it.
- **Champion Creator**: the child designs it from high-quality pre-made parts (bodies, faces, colours) and names it. Design continues all year — each project adds a visible part.
- Cosmetic parts are pre-made (to keep the art quality high); task content is always the child's own.

Coupling is loose. The champion is a small data file (a "Champion Profile"). Each project writes one ability into it; nothing depends on anything else. The "AI City" is a showcase that reads the profile and runs each ability in its own district, side by side — break one, the rest still work. Stored locally, plus an exportable "Champion Card" (nickname only, no personal data).

The City itself — the districts, the items to sort, the streets, the buses — is **pre-made scaffolding we build**. The child supplies only the *ability* that performs in it. So "show off" means: the child's trained ability runs a real job on our stage (their Eye sorts the city's recycling; their Legs clean a street). Each project ships with its own district scene, so this payoff works from project 1; the districts join into one full City at the finale.

Milestones: a short celebration after each cluster, and a year-end finale where the finished champion tours the whole City.

4. The AI buddy

Built on **opencode** (an open-source, provider-agnostic coding agent), locked down and wrapped in our own kid UI.

What it is: a named companion across all 17 projects. It tutors the concept, coaches the real-world job, teaches how to use generative AI, and — with the child's approval — helps tune the champion. Its tone matures from "here's how" early on to "what would you try?" by year-end.

Freedom, per axis:

Axis	Setting
Reading	High — sees the full module state and history
Acting	Only a whitelisted set of config actions, each child-approved
Autonomy	Approval on every action by default
Conversation	On-topic, with a kid-safety filter

Key rule: the buddy tunes config; it never supplies content or does the work.

Safeguards (enforced in config, not just prompts):

- Native tools (file, shell, web) are disabled. The buddy can only call our whitelisted config actions (e.g. `addSamples`, `setThreshold`, `trainAndEvaluate`).
- A hard, code-level gate checks every call.
- Kid-safety content filter on all chat and any generated output; bounded, teacher-keyed generation.
- Every action is previewed, approved, logged (teacher-visible), and undoable.

Branding: the child never sees opencode. They see our app and a buddy they named (e.g. "Sparky"). Our UI owns 100% of the look; opencode is the invisible engine. The buddy's personality is a prompt we write, with

the child's name and champion state added each turn.

Model: provider-agnostic and set by the school/teacher, not the child — a cloud model or a local one, swappable by config.

5. Tablet feasibility

Target hardware is entry-level tablets, which can run neither opencode nor a capable local model. So the buddy's engine runs on a server off the tablet; the tablet is always a browser client. This works on any tablet OS (iPad, Android, Chromebook). The games stay offline; the buddy is the one online feature.

tablet browser (our UI) $\rightleftharpoons$ opencode server (headless, locked down) $\rightleftharpoons$ model (cloud or local)

Open decision: where the server runs — cloud (needs internet) or a classroom box (offline in school).

Recommendation: prove it with a small L1 slice on a real tablet before deciding.

6. The 17 abilities

Ordered as a believable maturation: sense → act → plan → judge.

#	Champion part	The champion can now...	AI concept	City district
L1	Recycle-Eye (head)	name any object's material	train an image classifier	recycling sorter
L2	Deep-Scan Sense (chest)	find hidden hazards in noisy signals	speed/accuracy tuning + classification	building-site survey
L11	Watch-Sense (eye+)	flag real hazards without false alarms	detection threshold	intersection safety
L14	Sky-Sense (antenna)	spot real collision risks among many	clustering + sensitivity	air-traffic safety
L16	Mood-Sense (audio)	read feedback and gauge mood	sentiment + routing	sentiment board
L3	Rover-Legs (legs)	cover an area fully, no repeats	complete-coverage pathfinding	street cleaning
L4	Reflex-Wiring (nerves)	react to events with if-then reflexes	conditional logic	obstacle handling
L5	Twin-Drones (shoulder)	command helpers and split work	multi-agent allocation	mini fleet
L6	Haul-Planner (back)	plan an efficient loop under limits	route optimization	delivery routes
L7	Transit-Brain (core)	schedule a network to cut waiting	transfer timing	bus timetable
L8	Grow-Planner (core)	build ahead of forecast demand	forecast-driven placement	city planning
L9	Flow-Planner (core)	shift load, hold a reserve	load-shifting schedule	water/energy timing
L10	Grid-Warden (core)	govern AI agents with guardrails	oversight of autonomous agents	power market
L13	Ripple-Warden (core)	balance a system, watch side effects	second-order effects	traffic flow
L12	Judgment-Core (heart)	decide by values it learned from you	preference learning	ethical calls
L15	Cost-Sense (heart)	pick the right AI tool on a budget	model selection + gen-AI literacy	AI services
L17	Auditor-Core (heart)	oversee an imperfect AI, catch errors	human-in-the-loop	finance auditor

Two shared engines cover 9 of the 17: a **Model engine** (improve a model with data / a threshold; measure on unseen input) for the Senses, and a **Policy engine** (author a strategy; run it; score it) for the Body. The rest reuse pieces of both.

7. Build order

1. **L1 (Recycle-Eye) first** — proves the whole pattern end to end (data capture, tuning, the champion write, the full buddy loop).
2. Extract the two shared engines once a second project needs each.

3. Champion Profile + Card module (small, shared).
4. Buddy harness (locked-down opencode + our UI + safeguards).
5. Fan out the rest; the City grows one district per project.
6. Each project is independently verified — automated tests, accessibility on a real tablet, and an educator playtest — before shipping.

8. Open decisions

1. **"Show off" beat:** private payoff only (the child watches their ability run in the City), or add a social class-showcase at milestones.
2. **Sim-heavy projects (L2, L11, L14, L13):** the child tunes *our* scenarios (simplest), or also arranges the challenge first.
3. **Real surveys (L7, L9):** the child runs a light anonymized class/home survey to generate demand, or stay fully in-sim.
4. **Server location:** cloud (needs internet) vs. classroom box (offline) — recommend deciding via the L1 slice.
5. **Buddy "trust mode" default:** approve every action, or auto-apply trivially-reversible ones.
6. **Success thresholds** ($\geq 90\%$, etc.): placeholders, to be tuned per project during the build.
7. **Camera-first with no safety net** (the child re-collects, the buddy coaches) — currently accepted.

Appendix — decisions settled

- Project = configure/tune pre-built modules, no free-form code.
- Student content first: no pre-made content; the child owns all real content; the buddy guides but never supplies it.
- Champion = a humanoid the child designs; abilities accrete as visible parts; loose coupling via an append-only Champion Profile.
- Buddy = a named companion built on a locked-down opencode (native tools off, whitelisted actions, approve-to-act), branded entirely by our UI, with a provider-agnostic model set by the operator.
- Capstone: Champion Reveal Day — the child finalizes the design and presents the champion and its 17 abilities.
- All 17 abilities defined (section 6).